

# **GARBAGE COLLECTION IN JAVA**



## What is Garbage Collection?

Garbage Collection is process of reclaiming the runtime unused memory automatically. In other words, it is a way to destroy the unused objects.

To do so, we were using `free()` function in C language and `delete()` in C++. But, in java it is performed automatically. So, java provides better memory management.

## Advantages

- It makes java memory efficient because garbage collector removes the unreferenced objects from heap memory.
- It is automatically done by the garbage collector(a part of JVM) so we don't need to make extra efforts.

## How can an object be unreferenced?

There are many ways:

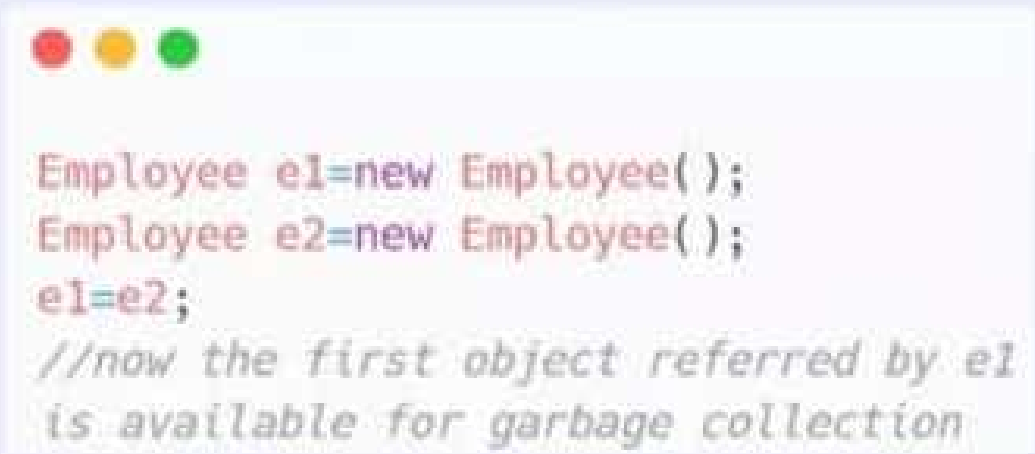
- By nulling the reference
- By assigning a reference to another
- By anonymous object etc.

## 1) By nulling a reference



```
Employee e=new Employee();  
e=null;
```

## 2) By assigning a reference to another



```
Employee e1=new Employee();  
Employee e2=new Employee();  
e1=e2;  
//now the first object referred by e1  
is available for garbage collection
```

## 3) By anonymous object



```
new Employee();
```

## finalize() method

The finalize() method is invoked each time before the object is garbage collected. This method can be used to perform cleanup processing. This method is defined in Object class as



```
protected void finalize(){} 
```

## gc() method

The gc() method is used to invoke the garbage collector to perform cleanup processing. The gc() is found in System and Runtime classes.



```
public static void gc(){}
```